Modeling the Migration of Dictyostelium Discoideum Cells in Gelatin Methacrylate Hydrogels In Order to Replicate Extracellular Matrix

Introduction

Cells don't live on flat surfaces; they operate in the complex 3D landscapes of the human body. Yet most studies still behave as if they do. This study focuses on understanding how living cells move and interact within 3D environments using custom-made hydrogels that mimic real human tissue. The problem being addressed is that 2D systems oversimplify how cells behave, limiting our true understanding of wound healing and tissue repair. This must be addressed because building accurate models of cell movement facilitates the creation of more effective biomaterials and regenerative treatments. This project will design materials that don't just contain cells but truly reflect how life operates in three dimensions.

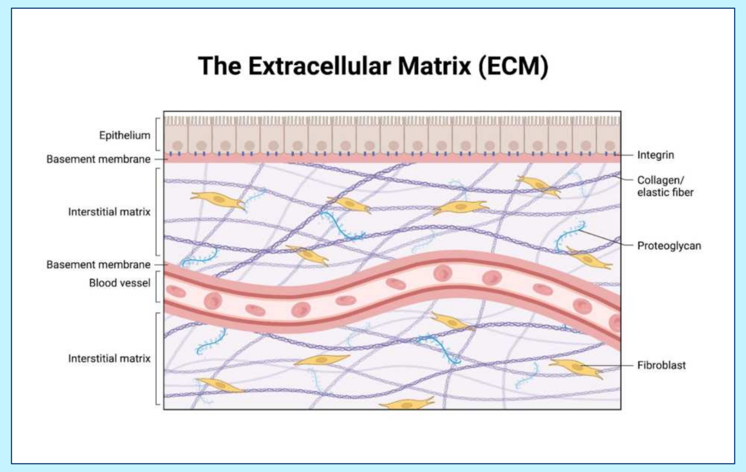

Image 1: ECM diagram

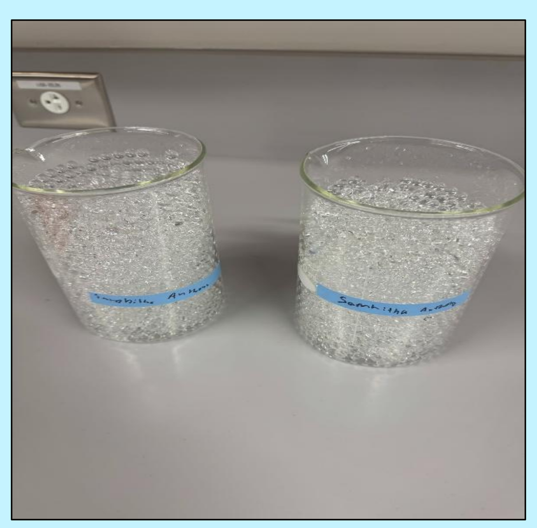

Image 2: Displays pre-made hydrogels before molding

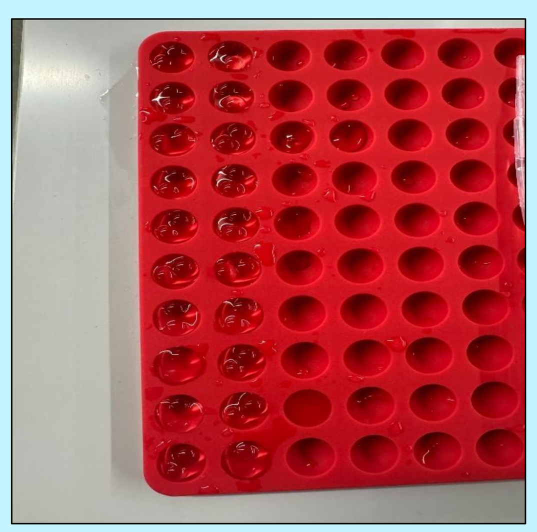

Image 3: shows the Pre-made hydrogels within a mold

Purpose

Understanding how cells move within 3D environments is essential for applications in regenerative medicine. The ability for cells to navigate 3D environments determines whether they can repair and regenerate themselves.

2D systems lack the complexity to represent the real body. Hydrogels are readily modifiable, water-rich polymers that provide an ideal platform for modeling the ECM (the network of molecules that provide support and signaling for cells). The outcomes of this project will create a safe, scalable platform for training in bioengineering research.

Methods

Setup Phase

1. Hydrogel Preparation

Different proportions of gelatin and photoinitiators were prepared, and photo-crosslinked

2. Cell Culture Preparation

Medium was prepared, and Dictyostelium discoideum was prepared.

Combining Phase

3. Embedding

Sterile pipettes were used to embed cells into solutions before photo-crosslinking

4. Surface Seeding

Cells were placed on top of already prepared hydrogels with a sterile pipette

Data Collection Phase

5. Imaging

The cells were imaged every 20 minutes using a fluorescence microscope

6. Analysis

Using ImageJ software, the speed, direction, and distance traveled by the cells was tracked.

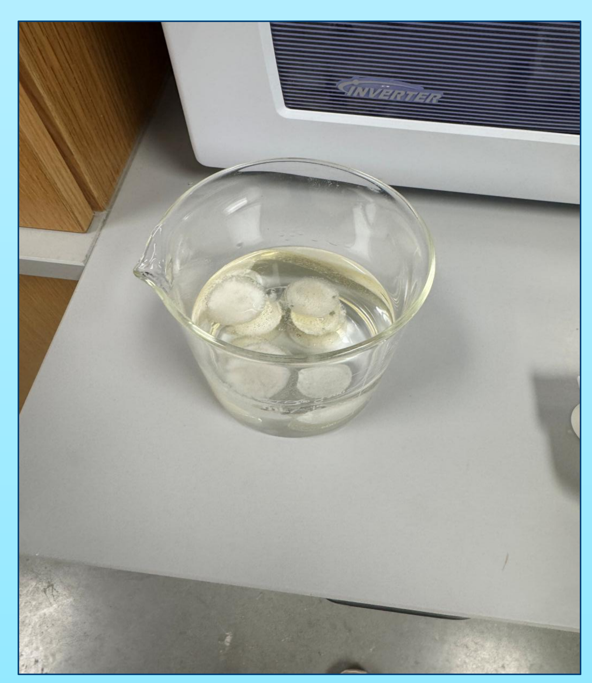

Image 4: Gels being subjected to a swelling test

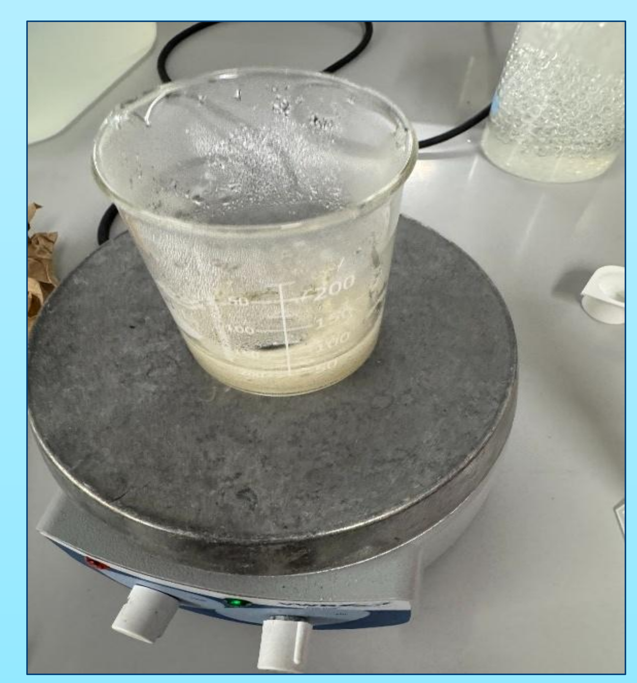

Image 5: Shows Hydrogels being prepared on a hot plate

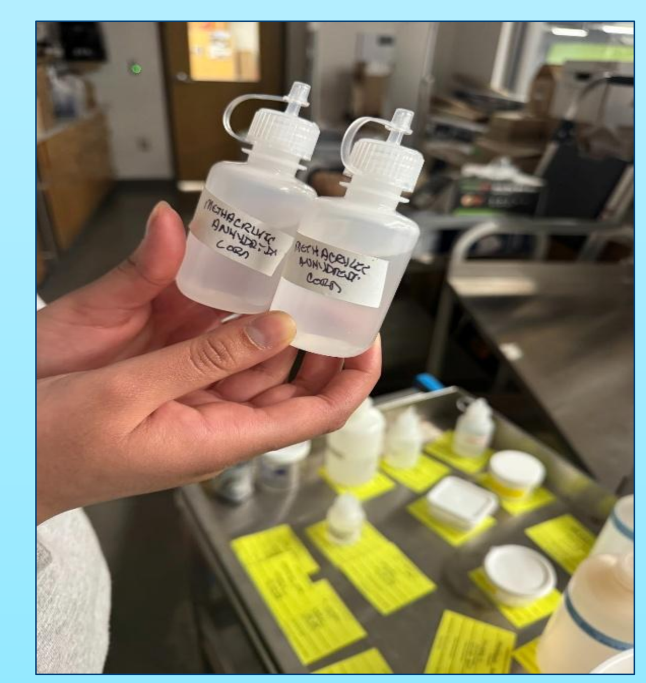

Image 6: displays the gelatin solution that was used

Results/Analysis

Salt concentratic	trial	initial mass	swollen mass	swelling ratio
0	1	9.8	16.4	0.67
	2	10.2	17	0.67
	3	9.9	16.2	0.64
0.10 M	1	10.5	15.8	0.5
	2	10	15	0.5
	3	10.3	15.4	0.5
0.30 M	1	11.2	14.8	0.32
	2	10.9	14.3	0.31
	3	11	14.5	0.32
0.60 M	1	12.1	14.2	0.17
	2	11.8	13.7	0.16
	3	12	13.9	0.16

Figure 7: Displays Data table displaying swelling ratio in water

H6

	A	B	C	D	E	F	G
1	Original mass	8.244	Original mass	10.348	Original mass	9.128	
2	Mass after 15 mi	8.768	Mass after 15 mi	10.892	Mass after 15 mi	9.297	
3	Mass after 20 mi	9.012	Mass after 20 mi	11.011	Mass after 20 mi	9.355	
4	Mass after 30 mi	9.251	Mass after 30 mi	11.202	Mass after 30 mi	9.782	
5	Mass after 45 mi	9.532	Mass after 45 mi	11.298	Mass after 45 mi	9.83	
6	Mass after 50 mi	9.599	Mass after 50 mi	11.35	Mass after 50 mi	9.892	
7	Mass after 55 mi	9.863	Mass after 55 mi	11.508	Mass after 55 mi	10.012	
8							
9							
10							
11							
12							

Figure 8: Data table displaying swelling ratio vs NaCl concentration

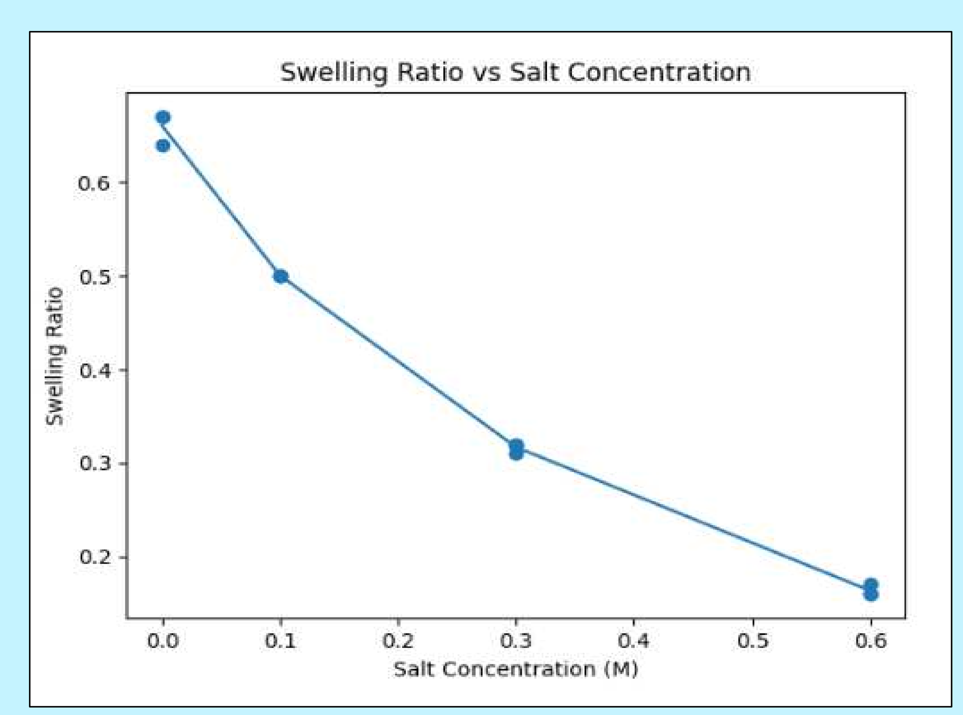

Figure 9: features a Graph displaying a data table of swelling ratio vs NaCl concentration

Conclusion

The importance of the physical and chemical properties of the surrounding environment on the migration of cells was highlighted in this study. Using time-lapse tracking, the movements were analyzed, and there was a clear distinction between the various types. From there, a comparison was drawn between the hydrogels and the human body's expected response. However, although there was some resemblance to how the ECM works, there was only a partial mimicking. There was not enough similarity to be able to claim that any of the hydrogels that were tested mimicked how the human body works.

Image 10: shows cells being separated in a centrifuge

Image 11: displays hydrogels being seeded with cells

The claim that can be made, however, is that among all types of hydrogels, gelatin methacrylate was the one that most closely replicated the human body because it was stiffer and had a lower photo-crosslinking density, which made its results more biologically relevant. Agarose gels had velocities of about 1.5–1.7 µm/min and experienced more directional movement. This is because the cells did not biologically interact with the structure, leading to the movement being more like a free fall, while GelMA hydrogels had velocities of about 0.5–0.7 µm/min due to being less porous and having more bonds for the cells to interact with.

Extension

Now that the initial data collection is over, there are several opportunities to do more research on this topic. The first and easiest extension of this project would be to just increase the sample size to be able to make more accurate claims. The next most logical step would be to include more chemical gradients for the cells to interact with, such as cAMP. This would make it easier to determine whether the cells are interacting with environmental biochemical properties or migrating randomly among other ECM components, such as collagen and fibronectin, to introduce more signals.

References

Katoh, M., et al. "Dictyostelium discoideum as a Non-Mammalian Biomedical Model." *PMC*, 2020.https://pmc.ncbi.nlm.nih.gov/articles/

Stewart, M., et al. "Dictyostelium discoideum: A Model for Investigating Genes Implicated in Neurodegeneration." *Frontiers in Cellular Neuroscience*, 2021.https://www.frontiersin.org/journals/cellular-neuroscience/articles/10.3389/fncel.2021.7595

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